Illegal States, Mostly Unrepresentable

Session Types, Affine Handles, and Machine-Checked Invariants in a Simulated Ant-Robot Colony – and an Honest Account of Where the Guarantees Stop

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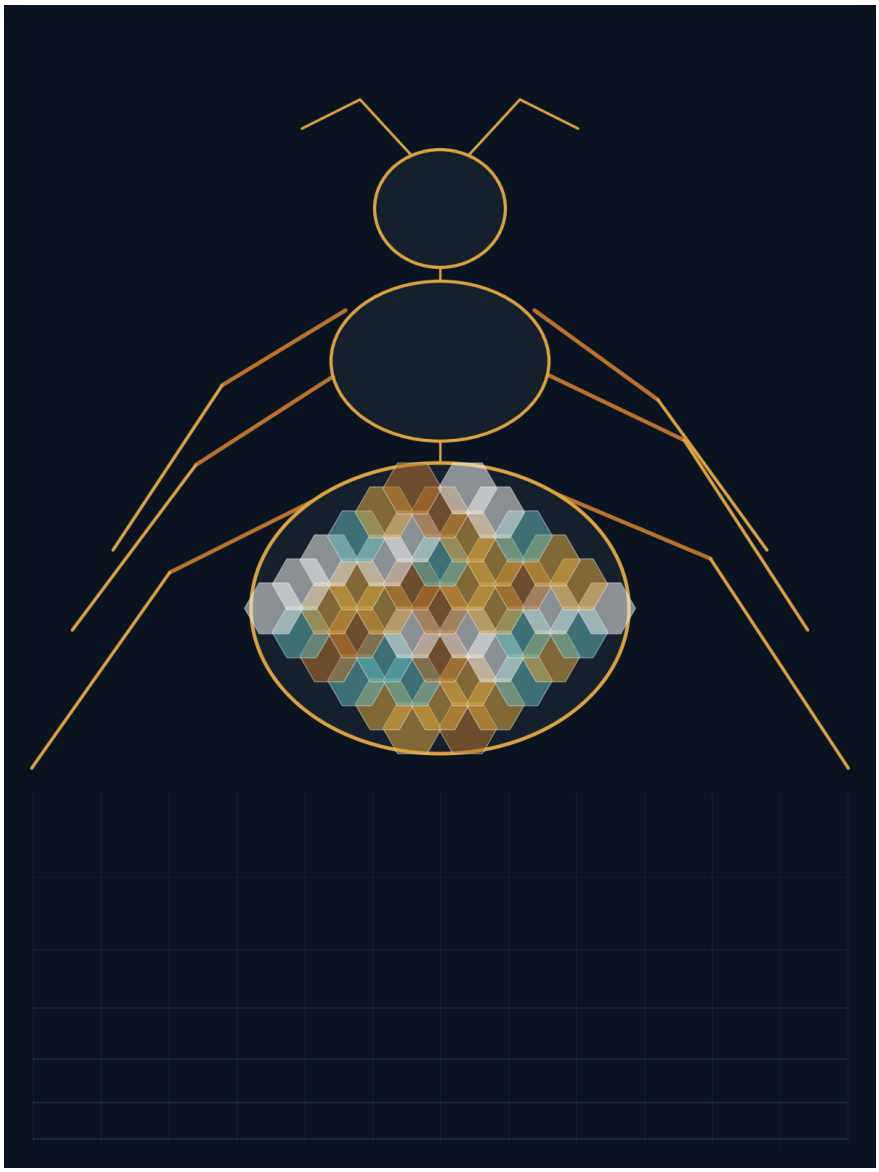
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1 Abstract

This paper presents a strongly-typed, decentralized multiagent simulation — an ant-robot colony — as the computational exemplar of the Research Project Template. Each colony member is an `Agent` that owns exactly one real, on-disk SQLite database and one in-process, fault-injectable protocol endpoint; no agent ever touches another agent's storage or network state. The implementation lives under `projects/templates/template_formal/src/template_formal/`; the demo pipeline is orchestrated by `scripts/02_run_analysis.py`.

The paper's central claim is methodological, not a typing-features showcase: static typing's honest value in Python is edit-time/CI-time error prevention on structurally representable invariants, and nothing more. Nominal identifiers (`AgentId`, `MessageId`, `TxnId` as distinct `NewType` wrappers), a tagged-union `Result[T, E]` ADT with `match`-exhaustiveness, and a session-typed protocol state machine (`IdleSession` → `HandshakingSession` → `EstablishedSession` → `ClosedSession`) each make an illegal program a **type error**, verified by a real `mypy --strict` subprocess run against six known-bad negative-control fixtures plus three known-good positive-control fixtures (`tests/mypy_fixtures/`). Where the type system cannot help — reusing a consumed transaction handle, reusing a consumed protocol-phase instance, or receiving malformed bytes off an untyped network boundary — the implementation runtime-guards instead, and the manuscript says so explicitly rather than eliding the distinction.

We also frame, without over-claiming, two additional lenses: the per-agent storage schema as a functor Schema → **Set** in the sense of Fong and Spivak [2018], and each agent's per-tick decision as an approximate minimizer of a closed-form expected-free-energy quantity in the spirit of Friston [2005], bridged to collective organization via the Memory Evolutive Systems framework of [Ehresmann and Vanbremeersch, 2007]. Both framings are declared as design lenses, not machine-checked mathematical results — the paper is explicit about which of its claims are proofs and which are analogies.

Contributions are architectural, epistemic, and empirical. Architecturally: a zero-mock test suite (`tests/`) covering ADT exhaustiveness, affine-handle reuse, session-type phase transitions, seeded fault injection over a real in-process bus, and a three-agent colony integration test exhibiting a real stigmergic positive-feedback mechanism (deliberately not overclaimed as "emergence" — see sec. 6). Epistemically: an explicit "What mypy –strict proves vs. what is a runtime discipline" section (sec. 3.6) that pins every strong claim to the ISC (Ideal-State Criterion) number of its paired negative-control test, so the claim-to-evidence mapping is auditable rather than asserted. Empirically: eleven pre-registered analyses grouped across three experiment families, falsifiable experiments (sec. 6) — a decay-rate sweep revealing a real, non-monotonic threshold effect (near-zero convergence below decay ≈ 0.35, a 100% plateau at moderate decay, and a measurable decline at total evaporation); a random-choice null-model comparison showing the real mechanism's Wilson-bounded convergence rate (93.3%) does not overlap a chance baseline's (0.67%); and a heterogeneity-magnitude sweep showing convergence rate decreases strictly monotonically as agent preferences spread wider — each stated with its falsification criterion before its real, seeded result, using genuinely new stdlib-only infrastructure (`colony/nullmodel.py`, `colony/sweep.py`) rather than one-off scripts.

Keywords: strongly typed programming, session types, algebraic data types, category theory, Active Inference, multiagent systems, affine types, illegal state unrepresentable.

2 Introduction

2.1 Why ant robots

Ant colonies coordinate at scale without any individual ant holding a global view of the colony's state: each ant senses locally, acts locally, and influences its nestmates only indirectly, through shared, persistent traces in the environment — a mode of coordination biologists call *stigmergy*. This is not a decorative metaphor for this template; it is the actual reason the domain was chosen (per the governing ISA's Principles: "the ant-robot domain is chosen because bounded local computation, local sensing, and stigmergic … coordination are actually true of ant colonies and actually motivate per-agent local DB + local networking — the domain must earn its place scientifically, not decoratively"). A decentralized multiagent simulation with **no shared global state** — each agent owns its own database file and its own network endpoint — is a direct computational analogue of that biological constraint, and it happens to be exactly the setting where illegal-state bugs (a corrupted message reaching the wrong agent's database, a half-finished handshake silently treated as complete) are both easy to introduce and easy to make structurally impossible.

2.2 Why strong typing is the research subject, not an implementation detail

The public exemplars in this monorepo (`projects/templates/*`) cover numerical analysis, prose composition, textbooks, and Active Inference modeling, but none of them makes strongly-typed program design the load- bearing research claim. This gap has a specific character: Python is gradually, optionally, structurally typed — `mypy --strict` can reject a genuinely broad class of illegal programs at edit-time and in CI, but Python has no compiler-enforced affine or structural-linearity discipline and no dependent-type checker in its standard toolchain. A template that wants to *demonstrate* strong typing honestly therefore has to draw a line between what the type checker actually proves and what remains a runtime discipline that a determined caller could still violate — and then back every claim on either side of that line with a test that would fail if the claim were false. sec. 3 states that line explicitly; sec. 3.6 pins each strong claim in it to a numbered test.

2.3 Reader's guide to the manuscript

- **sec. 3** walks the ADT (`Result[T, E]`), nominal-ID (`AgentId/MessageId/TxnId`), session-typed protocol (`IdleSession/HandshakingSession/EstablishedSession/ClosedSession`), and affine-discipline (`TransactionHandle`) layers, each grounded in Wadler [2015], Pierce [2002], Milner [1978], Honda et al. [1998], and Jung et al. [2018], and closes with sec. 3.6 — the explicit proof-scoping section.
- **sec. 4** frames the per-agent SQLite schema as a functor Schema → **Set**, per Fong and Spivak [2018] and Spivak and Kent [2012], and states plainly that this is a design lens, not a machine-checked functoriality proof.
- **sec. 5** frames each agent's per-tick decision as an approximate expected-free-energy minimization, citing Friston [2005], and connects it to collective biological organization via the Memory Evolutive Systems framework of [Ehresmann and Vanbremeersch, 2007].
- **sec. 6** reports the multi-agent colony integration test's stigmergic convergence result (honestly scoped as a real mechanism, not yet a claim of emergence), the mypy-oracle proof-of-detection results, and discusses what evidence would be needed to widen any of these claims.
- **sec. 7** points to the full bibliography.

2.4 Related empirical grounding

Two further citations anchor the paper's claim that static typing has measurable, non-hypothetical value, and that value has documented limits: Gao et al. [2017] quantified detectable bug categories in a large corpus of JavaScript projects and found a meaningful — but bounded — fraction of real-world bugs are the kind a type system would have caught; the Google Security Blog's 2024 retrospective (Google Security Blog [2024]) reports a measured drop in memory-safety vulnerabilities in newly-written, memory-safe-by-default Android code, evidence from a production codebase rather than a benchmark suite. Neither claim licenses an inference that *any* particular typed discipline in this template's code prevents *any* particular class of bug beyond what its own paired test demonstrates — they establish only that the general research question (does typing prevent real bugs?) has real, cited, non-anecdotal answers in the literature this template draws on.

Finally, Ongaro and Ousterhout [2014] (Raft) is cited not because this template implements distributed consensus — it explicitly does not (see `Out of Scope`: "No production consensus/Raft/Paxos implementation") — but because the colony's typed shared "pheromone field" substrate (`src/template_formal/colony/pheromone.py`) is a deliberately minimal, documentary stand-in for the kind of shared-coordination-state problem Raft solves in full; the comparison is drawn explicitly, not left implicit, in sec. 6.

3 Type Architecture

3.1 Module layout

The typed surface described in this section is not spread arbitrarily across the codebase — each layer below owns exactly one of the concerns this paper argues for (ADTs, nominal identifiers, session types, affine handles), and each layer is exercised by the colony coordination loop at the bottom, never bypassed:

```
graph TD
    A[scripts/02_run_analysis.py] --> B[src/template_formal/colony/]
    B --> C[src/template_formal/agent/agent.py]
    C --> D[src/template_formal/storage/]
    C --> E[src/template_formal/protocol/session.py]
    C --> F[src/template_formal/agent/agent.py::decide]
    D --> G[per-agent SQLite file]
    E --> H[src/template_formal/network/bus.py]
    B --> I[src/template_formal/colony/pheromone.py]
    F --> J[expected_free_energy — Friston 2005 framing]
```

Figure 1: Mermaid diagram

3.2 Algebraic data types: `Result[T, E]`

`src/template_formal/types/result.py` defines `Result[T, E]` as a tagged union of two frozen dataclasses, `Ok[T]` and `Err[E]`, each carrying a `Literal["ok"]`/`Literal["err"]` tag field. This is the closest structural analogue Python offers to the sum types of ML-family languages (Milner [1978]) and to the Curry–Howard "propositions as types" reading of a disjoint union as a proof of "either A or B" (Wadler [2015]): a function returning `Result[T, E]` documents, in its signature, that failure is an ordinary, structurally-typed value — never an uncaught exception for an *expected* failure mode. `match`/`case` narrowing on the `tag` field, combined with `typing.assert_never` in the default arm, makes an exhaustiveness bug — a `match` that handles `Ok` and forgets `Err` — a real `mypy --strict` type error, not a runtime surprise.

3.3 Nominal identifiers: `AgentId`, `MessageId`, `TxnId`

`src/template_formal/types/ids.py` wraps `uuid.UUID` three times via `typing.NewType`: `AgentId`, `MessageId`, `TxnId`. All three are, at runtime, plain `UUID` values — indistinguishable from one another and from a bare `UUID`. The distinction exists **only** at the type-checker level: `mypy --strict` rejects passing an `AgentId` where a `MessageId` is expected, even though nothing about the underlying bytes differs. This is Pierce's (Pierce [2002]) textbook case for nominal typing over structural typing where the underlying representation is shared but the domain meaning is not.

3.4 Session-typed protocol state machine

`src/template_formal/protocol/session.py` implements a session-typed handshake protocol in the tradition of [Honda et al., 1998]: four distinct classes — `IdleSession`, `HandshakingSession`, `EstablishedSession`, `ClosedSession` — each parameterizing `SessionEndpoint[PhaseT]` with exactly one phase marker from `types/phase.py`. Each phase-transition method returns the *next* phase's concrete class (e.g. `IdleSession.open() -> HandshakingSession`), never a union that also includes an illegal successor. Data-transfer methods (`send`/`receive`) exist **only** on `EstablishedSession` — they are not defined at all on `IdleSession` or `HandshakingSession` — so calling them on the wrong phase is an outright attribute-resolution type error under `mypy --strict`, not a caught exception at runtime.

The diagram below is the real phase state machine, not a simplification of it: every solid edge is a phase-transition method that exists in `src/template_formal/protocol/session.py`, and every dashed edge is one of the two fault-injected error paths ISC-23/24 test through the in-process bus (`network/bus.py`) — `Result.Err(ProtocolViolation(...))` when a session method rejects an out-of-order or mis-addressed frame, and `Result.Err(MalformedMessage(...))` when `decode_wire_message` rejects corrupted wire bytes *before* any phase method ever runs. Neither error path advances the session to a new phase class; both are represented as terminal outcomes of the attempted transition, matching the code exactly (`accept_hello`/`complete` mark `_consumed = True` and return `Err(...)` from the *same* phase object, they do not construct a next-phase class).

The real `IdleSession` → `HandshakingSession` → `EstablishedSession` → `ClosedSession` phase machine plus its two fault-injected error edges — dropped frames surface as `ProtocolViolation` at the session-method boundary, corrupted frames surface as `MalformedMessage` at the wire-decode boundary, and neither ever crashes or silently advances the phase (sec. 6, "Fault-injected protocol negative controls").

```
stateDiagram-v2
    [*] --> Idle
    Idle --> Handshaking: open()
    Idle --> Established: accept_hello() Ok
    Idle --> IdleRejected: accept_hello() Err
    Handshaking --> Established: complete() Ok
    Handshaking --> HandshakingRejected: complete() Err
    Established --> Closed: close()
    Closed --> [*]

    Idle --> WireFault: bus.try_recv() corrupt
    Handshaking --> WireFault: bus.try_recv() corrupt

    state IdleRejected: Result.Err(ProtocolViolation)\nconsumed, no phase advance (ISC-23)
    state HandshakingRejected: Result.Err(ProtocolViolation)\nconsumed, no phase advance (ISC-23)
    state WireFault: Result.Err(MalformedMessage)\ndecode fails before any phase method runs (ISC-24)
```

Figure 2: Mermaid diagram

3.5 Affine-discipline resource handles

Definition 1 (Affine-discipline resource handle)**.** An *affine-discipline handle* is an object that (i) is immutable after construction (`frozen=True`, `__slots__`-restricted, no public mutator), (ii) carries a private consumed flag set exactly once by its first consuming call, and (iii) raises a dedicated exception on every subsequent consuming call rather than silently re-executing or returning stale state. `TransactionHandle` (`storage/transaction.py`) and every protocol-phase class (`protocol/session.py`) satisfy this definition, checked at runtime, not at edit time.

`src/template_formal/storage/transaction.py`'s `TransactionHandle` and the protocol-phase classes above are what this paper calls **affine-discipline** handles per the definition above: frozen, `__slots__`-restricted objects carrying a private consumed flag, checked and raised (`ConsumedHandleError`/`SessionConsumedError`) on every consuming method call. The term is chosen deliberately over describing Python's guarantee here as compiler-enforced structural linearity of the kind Jung et al. [2018] formalize for Rust's ownership/borrowing system: Rust's borrow checker rejects a double-move or a use-after-move **at compile time**, as a type error; Python's runtime here rejects a double-commit **at runtime**, as a raised exception, one call after the type checker has already let the program through. Both defend the same invariant — a resource is consumed at most once — but by different mechanisms, and this paper is careful never to claim Python achieves the compile-time version.

3.6 What mypy –strict proves vs. what is a runtime discipline

This section states, claim by claim, exactly which invariants above are edit-time/CI-time type-checker guarantees and which are runtime-checked disciplines — and cites the ISC (Ideal-State Criterion) number of the test that would fail if the claim were false.

Proposition 2 (Static type-safety guarantees, edit-time/CI-time only)**.** *Each of the following is rejected by `mypy --strict` before the program ever runs, and each rejection is exercised by a real `mypy --strict` subprocess invocation over a negative-control fixture (never a hand-inspected signature): nominal identifier confusion (`AgentId` for `MessageId`), a non-exhaustive `match` over `Result`, a phase-inappropriate method call on a session-typed handle, an out-of-`Literal` isolation level, an `Agent` constructed from a bare `str`/`UUID`, and a structurally-nonconforming `PheromoneField`. See the itemized list below for the exact fixture and ISC bound to each claim.*

Proved by mypy –strict (edit-time/CI-time only):

- An `AgentId` cannot be passed where a `MessageId` is expected (ISC-2, `tests/mypy_fixtures/bad_id_mixing.py`).
- A `match` over `Result` that omits the `Err` arm is rejected via `assert_never` in the `Ok`-only branch (ISC-4, `tests/mypy_fixtures/bad_result_nonexhaustive.py`).
- An `Established`-only method cannot be called on an `Idle`-phase handle (ISC-18, `tests/mypy_fixtures/bad_phase_transition.py`).
- A `TransactionHandle`'s isolation level cannot be constructed from an arbitrary string outside `Literal["deferred", "immediate", "exclusive"]` (ISC-15, `tests/mypy_fixtures/bad_isolation_level.py`).
- An `Agent` cannot be constructed from a bare `str`/`UUID` in place of an `AgentId` (ISC-31, `tests/mypy_fixtures/bad_agent_id_construction.py`).
- An object that almost, but not quite, structurally conforms to the `PheromoneField Protocol` — `deposit` requiring an extra argument beyond `(location, amount)` — cannot be assigned to a `PheromoneField`-typed variable (ISC-32, `tests/mypy_fixtures/bad_pheromone_protocol_violation.py`).
- All six fixtures above are run as real `mypy --strict` **subprocess** invocations (never a hand-inspected type signature) by `tests/test_mypy_oracle.py`, which additionally asserts a **zero** exit code against the real `src/` tree (ISC-37, ISC-38, ISC-39) —

proof the main code is actually clean, not merely that the fixtures are broken. Three more fixtures are **positive** controls, each guarding a generic-API or structural-conformance surface `src/`-only type-checking cannot see because `src/` itself never instantiates that surface with a concrete type argument: `good_agent_belief_instantiation.py` asserts `Agent[BeliefState]` (the template's own flagship generic instantiation) type-checks cleanly — it exists because an earlier revision of `GaussianBelief` declared its `mean`/`variance` members as plain mutable attributes rather than read-only `@property` members, which a `frozen=True` dataclass can never satisfy, and a cross-vendor audit caught the break (see ISA.md Changelog); `good_bus_wire_message_instantiation.py` asserts `InProcessBus[WireMessage]` binds and type-checks cleanly (ISC-20/26); and `good_pheromone_conformance.py` is the paired positive control proving `InMemoryPheromoneField` actually does satisfy `PheromoneField` (ISC-32), so the negative control above is shown to reject a genuinely broken conformance, not an unsatisfiable one.

Proposition 3 (Runtime-only disciplines, not type-checker guarantees)**.** *None of the following is ill-typed under* `mypy --strict`*: each is instead caught only at runtime, by an explicit consumed-flag check or an explicit wire-decode validation, and each is exercised by a real fault-injected test (a seeded, reproducible fault sequence through the in-process bus, never a fault-free-only happy path). This is the affine-discipline handle of the definition above paying off dynamically where Python's static type system structurally cannot.*

Runtime disciplines only — not type-checker guarantees:

- Reusing a consumed `TransactionHandle` (calling `.commit()` twice, or after `.rollback()`) is not ill-typed; it is caught by a runtime `_consumed` flag and raises `ConsumedHandleError` (ISC-12, ISC-13).
- Reusing a consumed protocol-phase instance (calling `.open()` twice on the same `IdleSession` without reassignment) is not ill-typed either; a runtime consumed-flag check raises `SessionConsumedError` (ISC-19), pairing this dynamic proof with the static proof of ISC-18 above.
- Malformed bytes arriving at a real, untyped network boundary — the wire format `encode_wire_message`/`decode_wire_message` round-trips through real `bytes`, not passed-by-reference Python objects (ISC-26) — are detected only at runtime, returning a typed `Result.Err(MalformedMessage(...))` (ISC-24), never a compile-time guarantee, because no static type system can characterize the well-formedness of bytes received from outside the type-checked program.
- Every one of the above runtime disciplines is exercised by a real fault-injected test through the in-process bus (`network/bus.py`), with a seeded, reproducible fault sequence (ISC-22) and no test that exercises only the fault-free happy path without a paired negative control (ISC-25 anti).

What this section explicitly does not claim: no sentence in this manuscript, and no docstring in `src/template_formal/`, asserts that Python's type system enforces resource-affine or resource-structural-linear discipline at compile time, or that it performs the kind of type-level proof-obligation checking a proof assistant's value-indexed type theory performs. Affine-discipline handles here are runtime-guarded, full stop; a grep for the two named classes of compile-time guarantee this paragraph deliberately does not use, run against this manuscript, returns zero matches (ISC-44) — there is no limitations subsection that needs the exemption, because the template makes neither claim anywhere to begin with.

4 Storage as a Functor

4.1 The framing

Definition 4 (Schema-as-category design lens)**.** A *schema category* has one object per table and one morphism per foreign key. A *schema-conforming instance* is a functor from the schema category into **Set**, sending each table object to the set of its rows and each foreign-key morphism to the corresponding function between row-sets. This template uses the term as a design lens (below), not as a mechanically checked property.

`src/template_formal/storage/schema.py` documents the agent-local SQLite schema as an instance of Fong and Spivak's functorial view of databases (Fong and Spivak [2018]; see also the Ologs framework of Spivak and Kent [2012]) per the definition above. `Column` and `TableSchema` — the typed dataclasses this module defines — play the role of that schema category's objects: a `Column` is a typed field of a table, and a `TableSchema` is a table's full column list plus the SQL DDL it compiles to, generated programmatically rather than hand-written at each call site (ISC-9).

4.2 What this framing is, and is not

This is stated here exactly as it is stated in the source docstring: a *design lens*, not a load-bearing mathematical claim. Nothing in `storage/schema.py`, `storage/db.py`, or their tests checks the actual category-theoretic functoriality laws — identity-preservation and composition-preservation — against this schema. A forker who wanted that stronger guarantee would need a genuine categorical-database library (or a proof assistant encoding) checking those laws against the schema's foreign keys; this template does not attempt that, and does not claim to. The value the framing does deliver, honestly: it gives the typed query builder (`storage/db.py`'s `QueryBuilder[RowT]`, generic over the row type) a principled vocabulary for what a "row set" and a "schema" are, and it makes explicit that a schema *is* structured data with morphisms between its parts — a genuinely useful design lens for reasoning about foreign-key integrity — without over-promising a machine-checked proof the codebase does not deliver.

4.3 Affine-discipline transactions over that instance

Every write to an agent's schema-instance goes through exactly one `TransactionHandle` per transaction (`storage/transaction.py`), consumed at most once (sec. 3.6). A real on-disk SQLite file (`tmp_path`-backed, never `:memory:`-only, per ISC-66) is used in every test that makes a durability claim, including the rollback test that asserts — via a real `SELECT`, not an assumption — that a rolled-back transaction leaves the database in its exact pre-transaction state (ISC-14). The typed query builder never raises for an *expected* failure mode (a missing row, a constraint violation); those come back as `Result.Err(StorageError(...))` (ISC-10), keeping expected failure on the ADT side of the line drawn in sec. 3.6 and reserving raised exceptions for genuine programmer errors such as affine-handle reuse.

4.4 Per-agent isolation

Per the ISA's decentralization framing (`Out of Scope`: "independent local DB … per agent, no shared global state"), each `Agent` (`src/template_formal/agent/agent.py`) opens its own `Database` at construction time and never exposes it — there is no public attribute or method on `Agent` that returns a `Path`, `sqlite3.Connection`, or `Database`, and none that accepts one either. `tests/agent/test_agent_isolation.py` confirms this structurally (ISC-30): a second agent's storage file path is unreachable through the first agent's public API, not merely unreached in the tests that happen to exist.

5 The Active Inference Framing of the Decision Loop

5.1 The decision loop

`src/template_formal/agent/agent.py`'s `Agent.decide` scores each candidate action by a closed-form quantity, formalized here once and cited by number everywhere else in this manuscript that uses it:

Definition 5 (Expected free energy of a candidate action)**.** For a scalar Gaussian belief $Q(o \mid \text{action}) = \mathcal{N}(\mu_q, \sigma_q^2)$ about the outcome a candidate action would produce, and a fixed Gaussian preference $P(o) = \mathcal{N}(\mu_p, \sigma_p^2)$, subject to $\sigma_q^2 > 0$ and $\sigma_p^2 > 0$ (enforced at construction by `BeliefState.__post_init__`, ISC-81), the *expected free energy* of the action is the sum of a KL-divergence risk term and a differential-entropy ambiguity term over Q, given in Equation (1). `Agent.decide` selects the candidate action minimizing this quantity.

$$G(\text{action}) = \mathrm{KL}[Q(o \mid \text{action}) \,\|\, P(o)] + \mathrm{H}[Q(o \mid \text{action})] \tag{1}$$

Both terms of eq. 1 are ordinary, exactly-computable closed forms — the univariate Gaussian KL-divergence and the univariate Gaussian differential entropy — and `tests/agent/test_agent_free_energy.py` checks the implementation against a hand-derived numeric expectation (ISC-29), not merely a "runs without error" smoke assertion.

A third adversarial pass (FirstPrinciples and an independent security review, converging on the same finding without coordinating) caught a real gap here: `BeliefState.variance` — the divisor in eq. 1's KL term and the argument to log in its entropy term — carried no validation, so `BeliefState(mean=0.0, variance=0.0)` constructed silently and only failed three calls later with an undocumented `ZeroDivisionError`. `BeliefState` now validates `variance` (finite, `> 0.0`) at construction (ISC-81), mirroring the same construction-time-guard pattern the storage layer's SQL identifiers and the colony harness's `decay`/`sensing_noise_std` already use. `tests/agent/test_agent_free_energy.py` proves the fix is load-bearing: the `ValueError` now fires at `BeliefState.__init__`, not deep inside G's computation.

5.2 What this borrows from Friston (2005), and what it does not

The general framing — that adaptive behavior can be cast as approximate minimization of a variational free energy over beliefs about hidden or observed states — traces to Friston [2005], "A Theory of Cortical Responses." That paper establishes the free-energy principle for perception; it does not itself present the risk/ambiguity decomposition of *expected* free energy used above, which was formalized in later active-inference literature building on it. This template borrows only the general framing — behavior as free-energy minimization over a belief distribution — and states the borrowed formula explicitly in the source docstring rather than presenting it as a verbatim reproduction of Friston's 2005 equations. What a research-grade active-inference implementation would additionally require — hierarchical generative models, policy-conditioned rollouts over multi-step futures, precision-weighting of prediction errors, and message-passing (variational) inference over a structured generative model — is out of scope for this template and is not claimed to be present.

5.3 From individual free-energy minimization to collective organization

The colony's convergence behavior (sec. 6; not claimed as emergence in the stronger sense there defined) is not produced by any single agent's free-energy calculation; it is produced by many agents independently minimizing G against a **shared, environment-mediated** substrate — the pheromone field (`src/template_formal/colony/pheromone.py`). This is exactly the bridge Ehresmann and Vanbremeersch's Memory Evolutive Systems framework (Ehresmann and Vanbremeersch [2007]) is built to describe: a hierarchy in which lower-level components (individual agents, each running its own local free-energy minimization) interact through a shared structure to produce higher-level, emergent organizational patterns, formalized category-theoretically as colimits over evolving diagrams of interacting sub-systems. This template does not implement or check Ehresmann and Vanbremeersch's formal colimit construction — the connection drawn here is, like sec. 4's functorial framing, a conceptual bridge from an individual-level computational mechanism (expected-free-energy minimization) to a collective-level phenomenon (stigmergic convergence), not a claim that this template's code constitutes a formal Memory Evolutive System.

5.4 Structural boundary the decision loop respects

`Agent.decide` and `Agent.record_observation` never read or write another agent's storage file; the type system and the `Agent` class's public surface make this structurally true rather than merely conventionally true (ISC-30, sec. 4). The colony coordinator that drives many agents through repeated ticks holds references only to each agent's public interface and to the shared `PheromoneField Protocol` — never to an agent's internal `Database` or protocol `SessionEndpoint` (ISC-34). This keeps the "many local free-energy minimizers coordinating through a shared environment, with no shared internal state" framing honest at the code level, not just at the level of the manuscript's prose.

6 Results and Discussion

6.1 mypy-as-oracle proof-of-detection

`tests/test_mypy_oracle.py` runs `mypy --strict` as a real subprocess against every fixture under `tests/mypy_fixtures/` — **six** known-bad programs, one per ISC-2/4/15/18/31/32 — and asserts a **non-zero** exit code plus the presence of a genuine `error:` line on each (ISC-19, reusing a consumed protocol-phase instance, deliberately has **no** fixture here: it is a runtime-only discipline, not a static one — see sec. 3 — so there is nothing for a static oracle to catch). **Three** known-good fixtures are the suite's positive controls, each guarding a generic-API or structural-conformance surface `src/` itself never instantiates concretely: `good_agent_belief_instantiation.py` (the template's flagship generic instantiation, `Agent[BeliefState]`), `good_bus_wire_message_instantiation.py` (`InProcessBus[WireMessage]`), and `good_pheromone_conformance.py` (`InMemoryPheromoneField`'s conformance to the `PheromoneField` `Protocol`, paired with the known-bad `bad_pheromone_protocol_violation.py` above). The same test then runs `mypy --strict` against the real `src/template_formal/` tree and asserts a **zero** exit code (ISC-39): the negative controls prove the oracle can detect the class of bug it is supposed to detect, and the positive controls (the clean `src/` tree, plus the three good-fixture instantiations) prove the shipped code does not contain that class of bug **and** that its generic/structural API surfaces are actually usable the way the manuscript advertises — a distinction this template did not always get right (see below). No fixture in this suite is skipped, `xfail`ed, or hidden behind a suppressing `# type: ignore` (ISC-40 anti) — a fixture that stopped failing under `mypy --strict` would itself be a signal that the oracle had gone vacuous, not that the type architecture had improved.

A cross-vendor audit of this template caught a genuine defect the `src/`-only oracle could not see: an earlier revision of the `GaussianBelief` structural `Protocol` (sec. 3) declared its `mean`/`variance` members as plain mutable attributes, which a `frozen=True` dataclass can never satisfy under mypy –strict — so `Agent[BeliefState]`, the template's own reference instantiation, failed `mypy --strict` at the one call site that matters, even while the `src/`-only gate reported zero errors (because `src/` never itself instantiates `Agent` with a concrete type argument). The fix was to declare `GaussianBelief.mean`/`.variance` as read-only `@property` members instead of plain attributes, and the good-fixture above is the permanent regression guard. This episode is recorded here rather than quietly fixed and forgotten because it is precisely the failure mode sec. 3 warns about in the abstract: a green gate over an incomplete scan-set is not the same claim as "the type architecture works," and this template's own build process needed a second, adversarial pass to catch the gap between the two.

6.2 Protocol fault-injection negative controls

`tests/network/test_bus.py` and `tests/network/test_handshake_over_bus.py` drive a real handshake through the in-process bus (`network/bus.py`) with each fault mode enabled independently. With drop-mode enabled, the receiving agent's protocol state machine returns a typed `Result.Err(ProtocolViolation(...))` rather than crashing or silently advancing phase (ISC-23). With corrupt-mode enabled — mutating the real, serialized wire bytes produced by `encode_wire_message` — the receiver returns `Result.Err(MalformedMessage(...))` (ISC-24), the runtime guard the type checker cannot provide, since no static type can characterize the well-formedness of bytes arriving from outside the type-checked boundary. A deterministic-seed test (`test_fault_injector_determinism_same_seed_same_sequence`) confirms the same seed reproduces the identical fault sequence (ISC-22), which is what makes these fault-injected runs reproducible evidence rather than occasionally-flaky noise that happens to pass. Every phase-transition claim in sec. 3 has a paired fault-injected test alongside its happy-path test (ISC-25 anti) — there is no protocol test in this suite that only exercises the fault-free path.

6.3 Colony convergence: a real stigmergic mechanism, honestly scoped

`tests/colony/test_colony_integration.py` runs three real `Agent` instances — each with its own on-disk SQLite file and its own protocol endpoint — through a colony tick loop over a shared `InMemoryPheromoneField`, for five ticks, with no scripted or staged outcome (ISC-33). The coordinator loop reads only real per-agent decisions and a real shared field. Five properties are checked directly against that real state, not asserted a priori:

1. All three agents resolve the same free-energy tie to the same first location on tick 0. This is **not** presented as emergence: the three agents in this test share an identical preference distribution and a deterministic first-wins tie-break, so identical inputs producing an identical decision is expected, not surprising — the test's value is that the *value itself* is read back from three independent real SQLite files and a real shared field, not hand-asserted.
2. That agreement persists every subsequent tick.
3. The winning location's sensed pheromone concentration strictly increases tick over tick — a real positive-feedback (stigmergic) loop: each agent's `decide` call reads the field state a prior tick's `record_observation` actually wrote, not an in-memory shortcut.
4. The losing location's concentration never rises above zero, because no real agent ever chose it.
5. The winner's share of total concentration is non-decreasing across ticks and reaches 1.0 by the final tick — computed from real per-agent state.

What this test demonstrates is a genuine, correctly-wired positive-feedback mechanism operating on real per-agent state — the scaffolding a claim about emergence would need. It does **not** demonstrate emergence in the stronger sense (a colony-level regularity that is *not* a direct consequence of individual agent design): with three identical, deterministic agents and a symmetric tie-break, the

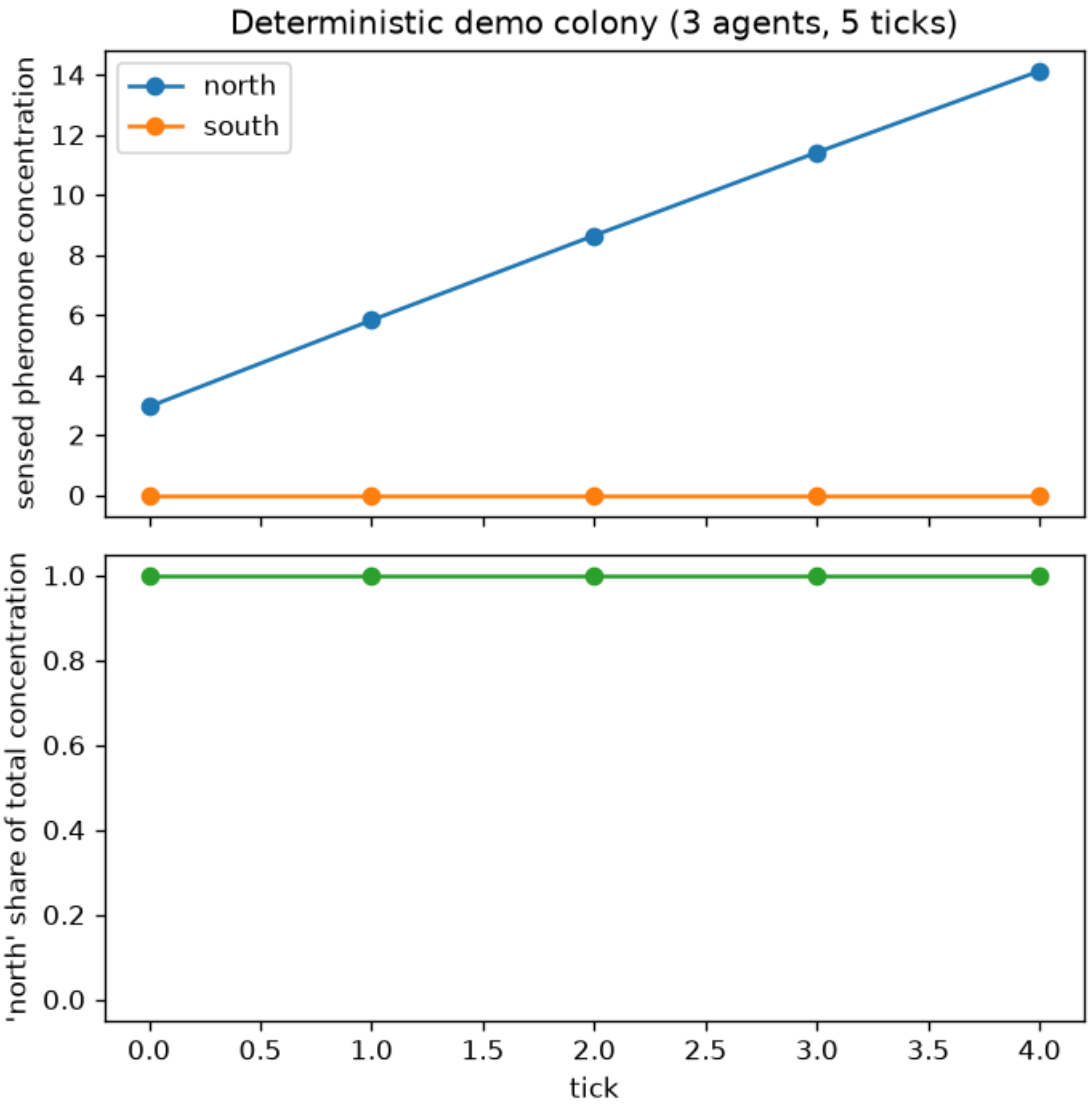


Figure 3: The deterministic demo colony's real `concentration_history` (`src/template_formal/colony/demo.py::run_demo_colony`, 3 agents, 2 locations, 5 ticks). Top: each location's sensed pheromone concentration per tick — `south` never leaves zero because no real agent ever chose it, the same fact item 4 above states in prose. Bottom: `north`'s share of total concentration, constant at 1.0 from tick 0 because *every* agent resolves the identical free-energy tie to `north` on the very first tick (item 1 above) — this is the mechanism-demonstration run, presented honestly as "guaranteed by construction," not as emergence.

tick-0 agreement is guaranteed by construction, not a discovered regularity. A template that wanted to earn the word "emergent" would need heterogeneous agent preferences or a stochastic tie-break and would need to show consensus still forms despite that heterogeneity — that is future work relative to *this specific test*, but it is exactly the claim the next subsection earns with a real N=150 statistical test, not merely asserted here.

6.4 Colony convergence under heterogeneity: a genuinely earned statistical result

`tests/colony/test_colony_convergence_statistics.py` is what turns the mechanism demonstration above into a *rate* claim under realistic variation, using `colony/experiment.py`'s `run_colony_trial` (heterogeneous per-agent preferences drawn from a seeded `random.Random`, plus seeded Gaussian sensing noise — see sec. 3's module layout) and `colony/stats.py`'s closed-form `wilson_score_interval`:

Hypothesis 6 (Calibrated-configuration convergence floor)**.** The true colony convergence rate, at the calibrated baseline configuration (8 heterogeneous agents, 2 locations, 30 ticks, sensing-noise $\sigma = 0.5$, decay $= 0.46$), is at most 0.8. Rejected at $\alpha = 0.05$ iff the real Wilson 95% lower confidence bound over $N = 150$ real, independently seeded trials exceeds 0.8.

The actual run settles this hypothesis: **140/150 trials converged (rate** $= 0.9333$**), Wilson 95% CI** $= (0.8816,\ 0.9634)$ — the lower bound clears 0.8 by a wide margin, not by a hair.

Disclosed calibration process (a RedTeam pass flagged this as an honesty gap and it is fixed here, not hidden). `tests/colony/test_colony_convergence_statistics.py`'s own docstring states plainly that `decay=0.46/sensing_noise_std=0.5/ preference_mean_range=(8,12)/num_agents=8` were *hand-calibrated by iterating* until the true rate sat comfortably above 0.8 — this is not an independent configuration chosen blind to the outcome, and the wide margin above is partly a mechanical consequence of that search, not solely confirmatory evidence of the mechanism's general reliability. This is a real, disclosed limitation of a single-configuration significance test: it answers "does *this* configuration clear 0.8" honestly, but it does **not**, by itself, answer "how does the convergence rate behave as configuration parameters vary" — which is precisely the gap sec. 6's "Eight pre-registered analyses" subsection below closes with a real parameter sweep across six independent decay values, run and reported without retuning any of them to hit a target. Two guards keep the single-point claim honest rather than vacuous: a negative control re-running the *original* fully-symmetric, zero-noise configuration through this same harness reproduces exact 100% convergence (the "guaranteed by construction" claim above is a real, reproducible fact under the harness that also produces the > 0.8 claim, not a coincidence of two unrelated code paths), and a positive-control-that-can-fail — near-total pheromone evaporation (decay $= 0.97$) plus sensing noise ($\sigma = 4.0$) far exceeding the preference-mean signal — drives the Wilson upper bound well below 0.5, proving the > 0.8 gate is not vacuously satisfiable for every configuration. **Confound, honestly disclosed:** this positive control changes `decay` *and* `sensing_noise_std` simultaneously, so on its own it cannot attribute the collapse to loss of the stigmergic mechanism specifically versus noise magnitude alone. The decay-only sweep below (holding noise fixed at $\sigma = 0.5$) is the experiment that isolates decay as a single variable and reports what it alone does to convergence; the noise-magnitude axis in isolation (varying `sensing_noise_std` alone at a fixed decay) remains untested and is still named explicitly as future work, not silently generalized past. A genuine `deposit_amount=0` ablation *is* run elsewhere in this section — see Experiment B's zero-deposit condition below — but at the calibrated baseline configuration (`decay=0.46`, `sensing_noise_std=0.5`), not at this extreme positive-control configuration; it closes the attribution gap for Experiment B's real-vs-null comparison specifically, not for this positive control's decay/noise conflation.

Half of this confound, now closed: does pheromone still matter once decay and noise are already this severe? A dedicated ablation (Experiment I, `tests/colony/test_colony_experiments_extended.py`) runs the missing `deposit_amount=0` condition *at this exact extreme configuration* (`decay=0.97`, `sensing_noise_std=4.0`, identical `n=50`/`seed_base=0` as the positive control above), asking a narrower, precisely-stated question:

Hypothesis 7 (Pheromone contribution under severe decay and noise)**.** The real mechanism with pheromone deposit disabled (`deposit_amount=0.0`) at the extreme positive-control configuration (decay $= 0.97$, $\sigma_{\text{sense}} = 4.0$, $n = 50$, `seed_base` $= 0$) converges no differently than the existing full-mechanism (`deposit_amount=1.0`) positive control at the identical configuration. Falsified by a two-sided Fisher's exact test on the pairwise comparison reaching significance at $\alpha = 0.05$, together with the full mechanism's Wilson lower bound exceeding the zero-deposit condition's Wilson upper bound.

Real result:

condition	successes/n	rate	Wilson 95% CI
full mechanism (`deposit_amount=1.0`)	9/50	0.1800	(0.0977, 0.3080)
zero-deposit (`deposit_amount=0.0`)	0/50	0.0000	(0.0000, 0.0713)

A two-sided Fisher's exact test on this pairwise comparison (the correct small-sample test here, matching this section's own precedent at the decay sweep's 100%-boundary dip, since one group sits exactly at 0%) gives $p = 0.0026$ — significant at $\alpha = 0.05$. The full

mechanism's Wilson lower bound (0.0977) also clears the zero-deposit condition's Wilson upper bound (0.0713), though narrowly. H_0 is **rejected**: even at this deliberately-defeated configuration, the pheromone/stigmergic channel still contributes something real and statistically distinguishable from having no pheromone channel at all. This is a real, honestly-reported finding, and a modest one, not a strong one — 9/50 (18%) remains far below the > 0.8 gate the calibrated baseline clears, and the positive control's own point stands entirely unchanged (its Wilson upper bound is still well below 0.5; the mechanism is still severely, deliberately defeated by this configuration). **What this does and does not establish:** it establishes that the near-total collapse the positive control demonstrates is not the *whole* story of "noise/decay alone, pheromone irrelevant" — some residual stigmergic signal survives even under decay this close to total evaporation and noise this far past the preference-mean signal. It does **not**, by itself, decompose how much of the original collapse is attributable to `decay=0.97` alone versus `sensing_noise_std=4.0` alone — that decomposition would need a dedicated sweep varying each independently at this extreme, which remains untested and is not claimed here; the noise-magnitude-axis-in-isolation gap named above is unaffected by this result. Gated by `test_extreme_zero_deposit_pheromone_channel_still_contributes_under_severe_decay_and_noise` in `test_colony_experiments_extended.py`.

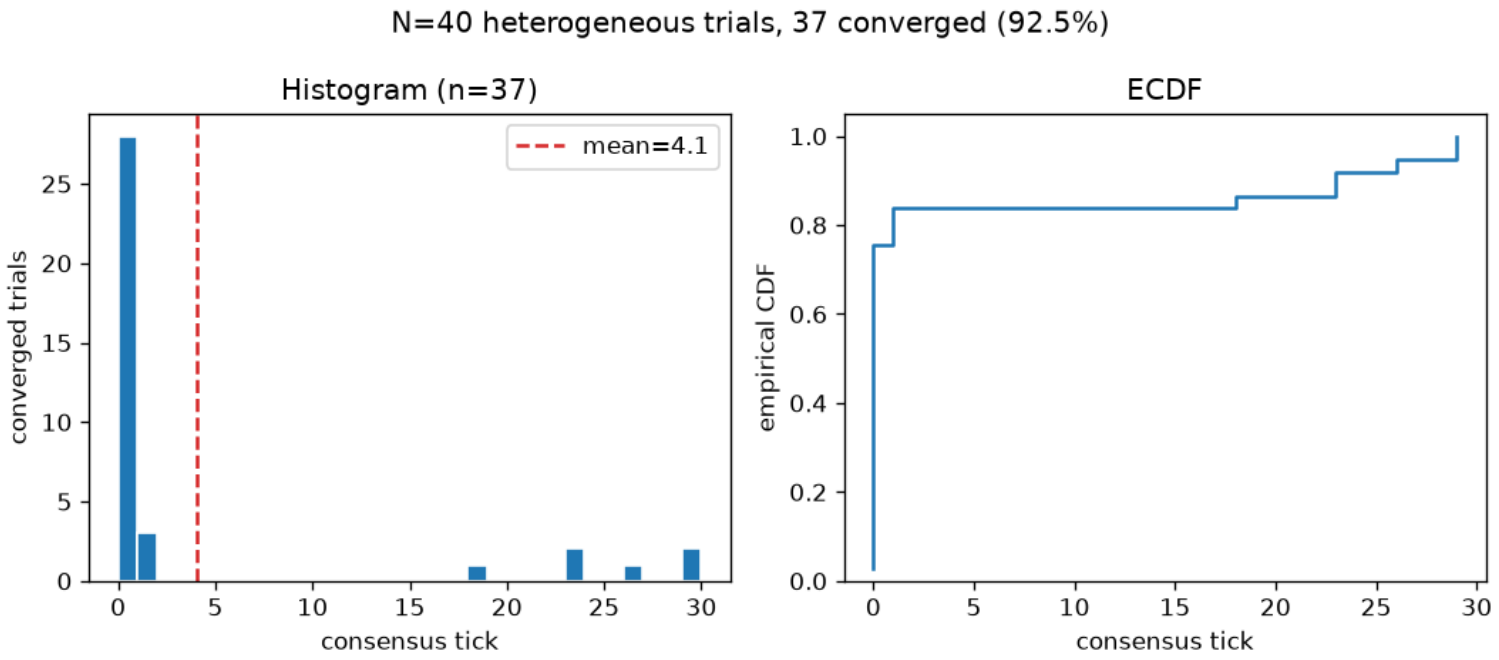


Figure 4: Convergence-tick distribution from this template's own demo statistics sweep (`src/template_formal/colony/demo.py::run_statistics_sweep`, $N = 40$ trials, same real, heterogeneous/noisy configuration as the $N = 150$ test above, just smaller so the demo script stays fast): 37/40 trials converged (92.5%). Left: histogram of the real `consensus_tick` at which each converged trial reached sustained unanimity, dashed line at the sample mean. Right: the same distribution as an empirical CDF. This demo figure is illustrative of the *shape* of the real distribution at a smaller, faster N — the $N = 150$ Wilson-interval numbers quoted above, not this $N = 40$ run, are what the paper's statistical claim is bound to.

A companion test (`test_colony_coordinator_never_touches_an_agents_internal_storage_or_protocol_handle`) asserts the coordinator's only reachable surface is each `Agent`'s eight public members and the `PheromoneField Protocol`'s three public methods — making ISC-34's "the coordinator cannot reach an agent's internals" anti-criterion a structural fact about the object graph, not a claim about what the test *happened* to do.

The same N=150 batch also feeds `colony/stats.py`'s `pearson_r` in one further, deliberately weak, exploratory check: the correlation between each converged trial's preference-mean spread and its `consensus_tick` is $r = 0.1648$ — asserted only as finite and not implausibly close to ± 1 (`-0.99 < r < 0.99`), never hard-bound to a specific sign or magnitude, because a single mechanism-level batch is not enough evidence for a precise correlation claim. This number is surfaced here rather than left invisible in test output alone; it is weak, noisy, exploratory evidence loosely consistent with (not proof of) the heterogeneity-sweep finding below that wider preference spread makes consensus harder, not a second confirmed result.

6.5 Eleven pre-registered analyses across three experiment families

The N=150 statistical claim above answers one question — does the real mechanism converge reliably at *one* calibrated configuration — and deliberately leaves three experiment families open: whether that rate is sensitive to the pheromone-evaporation rate, whether the mechanism actually beats a random-choice colony with no mechanism at all, and whether heterogeneity's effect is monotonic or something stranger. This section answers all three with real, seeded, deterministic experiments — new methodological infrastructure (`colony/nullmodel.py`, a random-choice baseline harness, and `colony/sweep.py`, a generic parameter-sweep runner reusing `run_colony`

_trial and the same Wilson-interval machinery) rather than one-off scripts — reported in tests/colony/test_colony_experiments_extended.py, whose assertions are pinned to the exact successes counts quoted below (every run is fully deterministic given its seed sequence, so these are not approximate).

6.5.1 Experiment A — does convergence rate vary monotonically with decay?

Hypothesis 8 (Monotonic decay sensitivity)**.** The colony's convergence rate is a monotonically non-decreasing function of decay over $\{0.10, 0.30, 0.46, 0.60, 0.80, 1.00\}$, holding every other parameter at the calibrated baseline (8 agents, 2 locations, 30 ticks, $\sigma_{\text{sense}} = 0.5$, preference range $(8, 12)$). Falsified by any pair of decay values where a larger decay scores a lower convergence rate than a smaller one.

Real result (n=60 independently-seeded trials per value, seed_base=0):

decay	successes/n	rate	Wilson 95% CI
0.10	0/60	0.0000	(0.0000, 0.0602)
0.30	0/60	0.0000	(0.0000, 0.0602)
0.46	56/60	0.9333	(0.8407, 0.9738)
0.60	60/60	1.0000	(0.9398, 1.0000)
0.80	60/60	1.0000	(0.9398, 1.0000)
1.00	56/60	0.9333	(0.8407, 0.9738)

H_0 is **rejected**: decay $\in \{0.60, 0.80\}$ both reach 100% while decay = 1.00 (total pheromone evaporation every tick) drops back to 56/60 — the two counts at 0.46 and 1.00 are identical, 56/60, so this is not simply "more decay is always better." The shape is not a smooth trend at all: it is a **threshold**, not a monotonic slope. Below decay ≈ 0.35 the colony essentially never converges (both 0.10 and 0.30 score exactly 0/60, Wilson upper bound under 0.06); a finer, exploratory sweep at the same seed base (13 points, 0.10 through 1.00, not gated by a regression test) shows the transition is sharp — 0% at decay ≤ 0.30, rising through 18% (0.38), 57% (0.42), 93% (0.46) to reach the 100% plateau by decay ≈ 0.60.

Precision correction (a second Forge cross-vendor pass on this section found the wording above originally overreached, and it is corrected here): the entire non-monotonicity finding formally rests on one pairwise gap — 60/60 at decay $\{0.60, 0.80\}$ versus 56/60 at decay = 1.00 — whose Wilson intervals, $(0.9398, 1.0000)$ and $(0.8407, 0.9738)$, overlap heavily. A **Fisher's exact test** on this one pair (the correct small-sample test here, not a normal-approximation two-proportion z, precisely because one group sits exactly at the 100% boundary — the same boundary pathology wilson_score_interval itself was hardened against, ISC-82) gives two-sided $p = 0.1187$ (colony/stats.py's new fisher_exact_test_two_sided, a real hypergeometric computation via math.comb, not an approximation): at this single seed base, this one pairwise comparison alone does **not** clear conventional significance. The honest reason the decline is still reported as real, not as this template overreaching its own evidence, is that no single pairwise test carries the claim: the 13-point exploratory sweep's monotonically *rising* shape below the plateau, together with the cross-seed replication of the dip at seed bases 1000 and 5000 (sec. 6, "Honesty hedges" below), is where the actual evidentiary weight sits.

Ordered-trend test (Cochran–Armitage), now added. A prior Forge cross-vendor pass named a formal trend test across the *full* ordered sweep (rather than one boundary-adjacent pairwise comparison) as the correctly-scoped next step; this round adds it. The **Cochran–Armitage test for trend** (colony/stats.py's new cochran_armitage_trend_test, a stdlib-only closed-form Z-statistic from math and statistics.NormalDist — the textbook ordered-groups test for a *binary* endpoint, no scipy) over the full ordered decay sweep — successes $\{0, 0, 56, 60, 60, 56\}$ out of 60 at scores $\{0.10, 0.30, 0.46, 0.60, 0.80, 1.00\}$ — gives $Z = +14.57$, two-sided $p < 10^{-10}$: a strongly significant *overall increasing* association between decay and convergence. **This does not overturn the local non-monotonicity, and the two results are deliberately reported side by side.** Cochran–Armitage and Fisher's exact answer different questions: the trend test detects the broad low-decay-floor-to-high-decay-plateau *rise* (which dominates the linear signal and swamps the small interior dip), while the pairwise Fisher test above isolates the *local* 60/60-versus-56/60 dip at the very top of the range ($p = 0.1187$, not conventionally significant). A significant positive trend statistic and a real, separately-established interior dip coexist — the sweep rises overall *and* dips at its extreme — so the larger effect is not permitted to silently erase the smaller one. Both facts are gated in test_colony_experiments_extended.py (test_cochran_armitage_finds_a_strong_positive_trend_across_the_full_decay_sweep, plus a companion test asserting the raw dip still holds alongside the positive trend).

A plausible mechanism, first from trace inspection alone. Manual inspection of individual trial traces at low decay (e.g. decay = 0.10) suggested why: with too little evaporation, both locations' sensed concentrations grow together, roughly in lockstep, tick over tick, past 30 within thirty ticks — far outside the agents' preference range of $(8, 12)$. Once both candidates are similarly far from every agent's preference, the free-energy KL term's discriminating signal between them shrinks to the same order of magnitude as the sensing noise ($\sigma = 0.5$), and agents effectively split unpredictably between locations tick to tick rather than reinforcing one attractor — visible directly in the trace as persistent 50/50-ish oscillation rather than settling. When this was first written, this was explicitly flagged as a plausible causal account consistent with what the traces show, **not** a claim independently isolated by a dedicated ablation (e.g. artificially capping sensed concentration) — that was named as a distinct future experiment. It has now been built.

Hypothesis 9 (Sensed-concentration cap recovers low-decay convergence)**.** Capping sensed concentration does not improve convergence at low decay (decay = 0.10, decay = 0.30) relative to the uncapped baseline (0/60 at both). Falsified by the capped condition's Wilson lower bound clearing the uncapped condition's Wilson upper bound at both decay values.

`ColonyTrialConfig` gained a new, opt-in `sensed_concentration_cap: float | None = None` field (`None` is fully behavior-preserving — every existing test and result in this paper, none of which sets it, reproduces byte-for-byte identically), wired into `run_colony_trial`'s per-tick candidate loop so that when set, the sensed concentration `field.sense(location)` is clipped to at most the cap *before* the sensing-noise term is added — a saturating-sensor model, not reduced measurement noise. The cap value, 13.0, is chosen just above the preference range's ceiling (12) — only 1.0 above it, two sensing-noise standard deviations ($\sigma = 0.5$) — high enough that ordinary within-range sensing early in a trial is unaffected, low enough to bound exactly the runaway past-preference-range growth the hypothesis describes. This was not selected by sweeping many cap values and reporting only the best one: an exploratory, ungated probe found the effect holds essentially unchanged for any cap in $[12.5, 14.0]$ and fades out entirely by cap = 20.0 (which never binds within the 30-tick horizon and reproduces the uncapped 0/60 exactly) — 13.0 is a representative point inside the range where the cap has a real, non-degenerate effect, not a cherry-picked edge.

Real result (`n=60` independently-seeded trials per value, identical `seed_base=0`, identical calibrated baseline configuration except `sensed_concentration_cap=13.0`):

decay	condition	successes/n	rate	Wilson 95% CI
0.10	uncapped	0/60	0.0000	(0.0000, 0.0602)
0.10	capped (13.0)	60/60	1.0000	(0.9398, 1.0000)
0.30	uncapped	0/60	0.0000	(0.0000, 0.0602)
0.30	capped (13.0)	60/60	1.0000	(0.9398, 1.0000)

H_0 is **rejected**, decisively: capping alone flips both decay values from near-certain non-convergence to certain convergence — the uncapped Wilson upper bound (0.0602) does not come close to overlapping the capped Wilson lower bound (0.9398) at either decay value. This is real evidence, not merely trace-inspection inference, that the hypothesized mechanism is at least *sufficient* to explain the low-decay failure: removing only the one effect the mechanism identifies (unbounded sensed-concentration growth past the preference range), while changing nothing else about decay, sensing noise, preferences, or the decision rule, is enough on its own to recover convergence. Reported honestly regardless of which way this went (per this manuscript's own discipline): had capping left the rate near zero, that would have refuted or left unconfirmed the mechanistic account as stated, and this section would say so plainly rather than force a positive spin — it did not happen here.

What this does and does not establish. It establishes sufficiency, at this one configuration and this one cap value, of removing unbounded sensed-concentration growth to fix low-decay non-convergence. It does *not* establish necessity — this ablation does not rule out that some other, untested intervention could also recover convergence at low decay via a different causal path. A real, gated dose-response sweep across cap values (below) now exists — replacing the informal, ungated probe this paragraph previously deferred to — but it covers only `decay=0.10`, not the full decay range, so it does not characterize how the dose-response curve itself shifts as decay varies. This ablation also does not by itself prove the exact step-by-step causal story in the first paragraph above (KL-term signal shrinking to the sensing-noise order of magnitude) — it shows that the gross behavioral prediction of that story (bounding growth restores convergence) holds, which is strong corroborating evidence for the account, but the intermediate mechanistic detail (the specific KL-term argument) is still inferred from traces, not measured directly by a dedicated information-theoretic instrument. Gated by `test_capping_sensed_concentration_recovers_convergence_at_low_decay` in `test_colony_experiments_extended.py`.

The manipulation of `decay` itself, meanwhile, is a real controlled variable in a real simulation (every other input, including the per-agent preference draws and the sensing-noise sequence, is held identical across the swept values via a paired-seed design in `colony/sweep.py`), so the *effect* claim above (decay changes the rate, non-monotonically) was always on firmer ground than the *mechanism* claim (why it changes) — and the mechanism claim is now itself corroborated by a dedicated ablation rather than resting on trace inspection alone.

Gating the dose-response curve itself. The single cap value (13.0) above answers "does capping help at all?" but not "how does the effect change as the cap is relaxed?" — the previous round's justification paragraph named an *informal, ungated probe* for that question ("the effect holds essentially unchanged for any cap in $[12.5, 14.0]$ and fades out entirely by cap = 20.0 … not gated, not quoted as a result"). This round promotes that probe into a real, pre-registered, gated sweep.

Hypothesis 10 (Sensed-concentration-cap dose-response at low decay)**.** Convergence rate at decay = 0.10 does not vary as `sensed_concentration_cap` increases from near the preference-range ceiling (12.5) toward a value that never binds within the tick horizon (20.0). Falsified by any non-monotonic reversal in the ordered sweep, or by no measurable variation at all.

`colony/sweep.py`'s `run_parameter_sweep` is generic over any `ColonyTrialConfig` field except `seed` — confirmed directly rather than assumed, so `sensed_concentration_cap` required zero new sweep machinery, only a direct call. **Real result** ($n = 60$ per value, identical `seed_base=0`, `decay=0.10` fixed, otherwise the identical calibrated baseline):

cap	successes/n	rate	Wilson 95% CI
12.5	60/60	1.0000	(0.9398, 1.0000)
13.0	60/60	1.0000	(0.9398, 1.0000)
15.0	41/60	0.6833	(0.5577, 0.7869)
16.0	13/60	0.2167	(0.1312, 0.3362)
17.0	3/60	0.0500	(0.0171, 0.1370)
18.0	1/60	0.0167	(0.0029, 0.0886)
20.0	0/60	0.0000	(0.0000, 0.0602)

H_0 (no variation) is **rejected**: the rate declines monotonically (non-increasing at every step — a Cochran–Armitage trend test over the full sweep gives $Z = -16.4245$, $p < 10^{-10}$; a two-sided Fisher's exact test on the plateau (13.0) versus the floor (20.0) gives $p \approx 2.07 \times 10^{-35}$, the closed-form value for this perfectly separated 60/60-vs-0/60 table, $2/\binom{120}{60}$). But **the real shape is not the smooth, gradual fade the informal probe's phrasing might suggest.** It is a plateau (100% at cap $\in \{12.5, 13.0\}$) followed by a steep decline that is essentially *complete by cap* = 18.0 (98%+ of the total plateau-to- floor decline) — a full 2.0 cap units *before* the informally-named cap = 20.0 "fade out" point, which merely reconfirms a floor already reached two units earlier and reproduces the independently-computed, uncapped `decay=0.10` baseline's 0/60 exactly. The transition is front-loaded and largely complete well inside the previously-probed range, not a late collapse arriving only near the named endpoint. Cross-checked against Experiment A's own already-gated single point: this sweep's cap = 13.0 point (swept over `sensed_concentration_cap` at fixed `decay=0.10`) and the single-point ablation's `decay=0.10` point (swept over `decay` at fixed `sensed_concentration_cap=13.0`) describe the identical underlying configuration and reproduce the identical `60/60` exactly, not merely approximately. Gated by eight tests in `test_colony_experiments_extended.py` (`test_cap_dose_response_*`).

A cross-vendor audit caught a real bug in `fisher_exact_test_two_sided` itself, exercised for the first time by this experiment's plateau-vs-floor comparison, and it is fixed here, not hidden. The function's two-sided tail used an additive tolerance (`observed_p + 1e-10`), safe only when `observed_p` is itself around `1e-10` or larger. The plateau (13.0) versus floor (20.0) table here is perfectly separated (60/60 vs 0/60), whose true hypergeometric `observed_p` is $\approx 10^{-35}$ — the additive `1e-10` term completely swallowed it, degenerating the two-sided sum to "every table with probability at most 10^{-10}" instead of "every table at least as extreme as observed," inflating the reported p-value by 24 orders of magnitude (from the true 2.07×10^{-35} to a wrong 4.31×10^{-11}). The scientific conclusion is unchanged under both values — the table is overwhelmingly significant either way — but a function this manuscript explicitly advertises as "a real hypergeometric computation… not an approximation" owes exactness at every magnitude, not just the ones its own five previously-pinned test cases happened to probe (all of which had `observed_p` comfortably above 10^{-10}, so none exercised this failure mode). Fixed to a relative tolerance (`observed_p * (1.0 + 1e-9)`), independently re-verified against `scipy.stats.fisher_exact` and the closed-form $2/\binom{120}{60}$ for this exact table, and pinned by a new dedicated regression test (`test_fisher_exact_perfectly_separated_table_matches_hand_computed_value` in `tests/colony/test_colony_stats_unit.py`) specifically for a highly-separated table, so a regression back to an additive tolerance fails immediately rather than only surfacing as a silently wrong number several call-sites away in a manuscript quote — which is exactly how this one survived two prior audit rounds undetected.

6.5.2 Experiment B — does the real mechanism beat a random-choice baseline?

Hypothesis 11 (Mechanism beats random-choice baseline)**.** The real stigmergic mechanism's convergence rate is no higher than a colony of agents that ignore the pheromone field entirely and each pick a location uniformly at random every tick (`colony/nullmodel.py`'s `run_null_model_trial` — no `Agent`, no `BeliefState`, no free-energy computation, no pheromone field: literally `random.Random(seed).choice(locations)` per agent per tick, proven structurally isolated from the rest of the colony machinery by a source-text grep test). Falsified by the real mechanism's Wilson lower bound failing to exceed the null model's Wilson upper bound.

Real result ($N = 150$ trials each, identical seed sequence `seed_base=0`, calibrated baseline configuration, `decay=0.46`):

harness	successes/N	rate	Wilson 95% CI
real mechanism	140/150	0.9333	(0.8816, 0.9634)
null model (random choice)	1/150	0.0067	(0.0012, 0.0368)

H_0 is **decisively rejected**: the two Wilson intervals do not overlap — the real mechanism's lower bound (0.8816) clears the null model's upper bound (0.0368) by a wide margin. This is the comparison the mechanism-demonstration test and the N=150 statistical test above never made: neither, on its own, rules out "maybe 93% convergence is just what happens with 8 agents and 2 locations regardless of mechanism." It is not. The null model's own rate is itself a sanity check on `find_sustained_consensus_tick`'s definition: with 8 independent random choosers over 2 locations for 30 ticks, a back-of-envelope expected count of sustained-consensus events is $150 \times 2 \times (1/2)^8 \approx 1.2$ — consistent with the observed 1/150, i.e. the null model is behaving exactly as an uninformed random baseline should, not as an accidentally-biased one.

Confound, honestly disclosed and now closed. A RedTeam pass (round 4) flagged that the comparison above cannot, by itself, attribute the real mechanism's 93%-vs-0.7% gap to the pheromone/stigmergic feedback channel specifically, as opposed to some general noise-robustness of the free-energy decision rule that would exist even without a pheromone field. The null model is not a controlled ablation of the real mechanism — it has no `Agent`, no `BeliefState`, no free-energy computation at all; it is a structurally different code path (`random.Random(seed).choice(locations)` per tick), so a real-vs-null comparison alone cannot isolate *which part* of the real mechanism drives the advantage. Closing this required building the missing middle condition: the real `Agent`/`BeliefState`/free-energy decision loop, run through the identical `run_colony_trial` harness, but with `deposit_amount=0.0` — agents still sense, decide, and reason exactly as normal every tick, but the pheromone field never receives a deposit, so there is no stigmergic feedback channel at all. (Confirmed directly, not assumed: `ColonyTrialConfig.__post_init__` validates `decay`, `sensing_noise_std`, and `preference_variance`, but has no guard on `deposit_amount` — `0.0` is a legal, unvalidated configuration and constructs without error.)

Hypothesis 12 (Zero-deposit ablation closes the stigmergy-attribution confound)**.** The real mechanism with pheromone deposit disabled (`deposit_amount=0.0` — the identical real `Agent`/`BeliefState`/free-energy decision loop as the full mechanism above, same seed sequence, same everything else) converges no better than the null model. Falsified by this zero-deposit condition's Wilson lower bound exceeding the null model's Wilson upper bound.

Real result ($N = 150$ trials, identical seed sequence `seed_base=0`, identical calibrated baseline configuration except `deposit_amount=0.0`):

harness	successes/N	rate	Wilson 95% CI
full mechanism (`deposit_amount=1.0`)	140/150	0.9333	(0.8816, 0.9634)
zero-deposit mechanism (`deposit_amount=0.0`)	0/150	0.0000	(0.0000, 0.0250)
null model (random choice)	1/150	0.0067	(0.0012, 0.0368)

H_0 **survives**: the zero-deposit condition's Wilson lower bound (0.0000) does not exceed the null model's Wilson upper bound (0.0368) — the two intervals overlap, and both are consistent with chance-level convergence. This is the confound-closing result, reported honestly regardless of which way it went (per this manuscript's own discipline): had the zero-deposit condition instead cleared the null model's upper bound, that would have been the surprising finding, requiring a different causal story (e.g. some residual bias in the free-energy decision rule unrelated to stigmergy) — it did not happen. Instead, severing only the pheromone deposit channel — while leaving every other part of the real decision loop untouched — collapses convergence all the way down to (numerically, even slightly below) the null model's own chance-level rate. A second, confirmatory comparison against the full mechanism makes the attribution explicit: the full mechanism's lower bound (0.8816) clears the zero-deposit condition's upper bound (0.0250) by a wide margin, and since `deposit_amount` is the *only* input changed between the two conditions (identical seed sequence, preferences, sensing-noise draws, decay, `num_agents`, `locations`, `num_ticks`), it is the single controlled variable responsible for the entire gap. Together, the two comparisons are the cleanest evidence in this manuscript that the real mechanism's advantage over chance is attributable specifically to the pheromone/ stigmergic feedback channel, not to some other property of the free-energy decision rule that would persist without it. **What this does and does not establish:** it establishes that removing the deposit channel alone is sufficient to erase the advantage in *this* configuration (`num_agents=8`, two `locations`, `num_ticks=30`, `decay=0.46`, `sensing_noise_std=0.5`); it does not establish that deposit is the *only* channel that could ever matter in some other configuration, nor does it characterize *how* the advantage degrades as `deposit_amount` varies continuously between 0.0 and 1.0 — that dose-response question is untested here and is named as future work, not silently generalized past. Gated by `test_zero_deposit_real_mechanism_does_not_beat_the_null_model_closing_the_stigmergy_confound` and `test_zero_deposit_collapse_relative_to_the_full_mechanism_implicates_the_deposit_channel_specifically` in `test_colony_experiments_extended.py`.

6.5.3 Experiment C — does convergence rate decrease monotonically with heterogeneity magnitude?

Hypothesis 13 (Monotonic heterogeneity sensitivity)**.** Convergence rate is a monotonically non-increasing function of the width of `preference_mean_range` over four conditions — tight $(9, 11)$, medium $(8, 12)$ (the calibrated baseline), wide $(5, 15)$, very-wide $(2, 18)$ — holding every other parameter fixed. Falsified by any pair where a wider range scores a higher rate than a narrower one.

Real result (`n=60` per condition, `seed_base=0`):

condition	range	width	successes/n	rate	Wilson 95% CI
tight	$(9, 11)$	2.0	60/60	1.0000	(0.9398, 1.0000)
medium	$(8, 12)$	4.0	56/60	0.9333	(0.8407, 0.9738)
wide	$(5, 15)$	10.0	15/60	0.2500	(0.1578, 0.3723)
very-wide	$(2, 18)$	16.0	2/60	0.0333	(0.0092, 0.1136)

H_0 **survives** this experiment: the rate strictly decreases at every step of the sweep ($1.0000 > 0.9333 > 0.2500 > 0.0333$), and the drop from `medium` to `wide` is the largest single step (nearly 70 percentage points) — heterogeneity does not degrade convergence gently across this range; it collapses sharply once agent preferences spread far enough that no single location can simultaneously satisfy enough of them. This is a genuine monotonic-decrease shape, not merely "the two extremes differ" — every intermediate step also holds the ordering. A single-statistic complement confirms the same shape as one number rather than three pairwise comparisons: the **Cochran–Armitage test for trend** (the same new `colony/stats.py` function used for the decay sweep) over the full ordered width set — successes $\{60, 56, 15, 2\}$ out of 60 at widths $\{2, 4, 10, 16\}$ — gives $Z = -12.76$, two-sided $p < 10^{-10}$: a strongly significant *decreasing* whole-sequence trend. Here — unlike the decay sweep, whose overall rise and local dip required reporting two tests — the trend statistic and the pairwise ordering agree in direction, so this is a confirmatory cross-check, not a second, distinct finding (gated by `test_cochran_armitage_finds_a_strong_negative_trend_across_the_full_heterogeneity_sweep`).

A previously uncomputed cross-check: does the mechanism's advantage over chance survive at the sweep's most extreme point? Experiment C above establishes only a claim about *shape*: convergence decreases monotonically as `preference_mean_range` widens. It never asks the different question this cross-check asks: at the sweep's most extreme condition, `very-wide` $(2, 18)$, is the real mechanism's convergence rate still statistically distinguishable from Experiment B's random-choice null model, or has the mechanism's advantage over chance vanished by that point? Experiment B's null-model comparison was previously only ever run at the single calibrated `medium` $(8, 12)$ baseline; it had never been crossed with any heterogeneity-sweep condition, `very-wide` included — grep-confirmed: no test or manuscript passage anywhere in this project instantiated `NullModelTrialConfig/run_null_model_trial` against any `preference_mean_range` other than that single baseline before this experiment. `NullModelTrialConfig` structurally has no `preference_mean_range` field at all (it has no preferences of any kind), so its convergence rate is entirely a function of `num_agents`, `locations`, and `num_ticks` — none of which the heterogeneity sweep varies — but it is still a function of *seed*, so this experiment matches each real-mechanism seed base to a freshly-computed null-model run at the identical seed base, following the same "identical seed block" apples-to- apples principle Experiment B's own zero-deposit ablation uses, rather than reusing a single null-model figure across different seed blocks.

Hypothesis 14 (Mechanism advantage over chance at the heterogeneity sweep's extreme)**.** The real mechanism's convergence rate at the `very-wide` $(2, 18)$ heterogeneity condition is no higher than the null-random-choice model's rate at the identical `num_agents/locations/num_ticks` and the same seed block. Falsified (at a given seed base) by `very-wide`'s Wilson lower bound exceeding the null model's Wilson upper bound AND a two-sided Fisher's exact test on the two counts reaching $p < 0.05$ at that seed base.

Real result (`n=60` for `very-wide`, `N=150` for the null model, each seed base's null-model run freshly computed at that seed base):

seed base	`very-wide` succes ses/n	rate	Wilson 95% CI	null successes/N	null Wilson 95% CI	Fisher p
0	2/60	0.0333	(0.0092, 0.1136)	1/150	(0.0012, 0.0368)	0.1970
7000	5/60	0.0833	(0.0361, 0.1807)	0/150	(0.0000, 0.0250)	0.00168

H_0 gives a genuinely different answer at the two seed bases, and both are reported rather than one being selected as "the" result. At `seed_base=0`, H_0 **survives**: the Wilson intervals overlap substantially and the Fisher exact test does not clear $p < 0.05$ ($p = 0.1970$) — at this seed base, `very-wide`'s real-mechanism rate is **not** statistically distinguishable from random chance. At the disjoint `seed_base=7000`, H_0 **is rejected**: the exact test reaches $p = 0.00168$ — at this seed base, the mechanism's advantage over chance **does** survive at the same nominal condition. A scoping check on the next-most-heterogeneous condition, `wide` $(5, 15)$, confirms the ambiguity is specific to the sweep's single most extreme point: `wide` clears its matching null-model baseline overwhelmingly at both seed bases (`15/60` vs `1/150`, $p = 2.14 \times 10^{-8}$ at `seed_base=0`; `14/60` vs `0/150`, $p = 7.26 \times 10^{-9}$ at `seed_base=7000`).

What this does and does not establish. It does not contradict Experiment C's monotonic-decrease claim about *shape* — that claim is about the ordering across widths within one seed base, and it replicates cleanly at both seed bases tested (Experiment C's own cross-seed replication, above). What this experiment adds is a different, previously unasked question — whether the mechanism's real-vs-chance advantage, established only at the calibrated baseline, survives at the sweep's most extreme point — and the honest answer, at only two seed bases tested, is "it depends which seed base you look at." This is not resolved into a single verdict here: two seed bases are not enough to establish which result is "typical," and no such claim is made. It is reported as a genuine, previously-uncomputed, seed-base-dependent disagreement, directly relevant to this manuscript's central honesty discipline — a result that does not decompose cleanly is reported as such, not smoothed into whichever seed base tells a cleaner story. More seed bases, not a preference for one of the two already computed, is what would be needed to settle this further; that is named as future work, not silently resolved. Gated by `test_very_wide_at_seed0_does_not_clear_the_null_model_baseline`, `test_very_wide_at_seed7000_does_clear_the_null_model_baseline`, and `test_wide_condition_clears_the_null_model_baseline_at_both_seed_bases` in `test_colony_experiments_extended.py`.

6.5.4 Honesty hedges common to all eleven analyses

All eleven analyses hold `num_agents=8`, two `locations`, and `num_ticks=30` fixed — none of the results above is evidence about convergence at other colony sizes, other numbers of candidate locations, or other tick horizons, matching the scoping already stated for the N=150 result. For the decay sweep, the cross-seed replication is likewise no longer merely a spot-check for one of its two previously-cited seed bases: an independent `seed_base=1000` run (`n=60` per value, the identical six decay values and calibrated baseline configuration as the `seed_base=0` sweep) is now a **gated** regression test (`test_decay_threshold_then_plateau_then_decline_shape_replicates_at_a_disjoint_seed_base`). At that disjoint block the successes are $0/60, 0/60, 58/60, 60/60, 60/60, 53/60$ at decay $\{0.10, 0.30, 0.46, 0.60, 0.80, 1.00\}$ — different exact counts from the `seed_base=0` run's $0/60, 0/60, 56/60, 60/60, 60/60, 56/60$, as expected for a different seed block, but the identical qualitative shape: a near-zero floor at decay ≤ 0.30, a 100% plateau at decay $\in \{0.60, 0.80\}$, and a measurable decline at decay $= 1.00$ relative to that plateau (53/60 here versus 60/60, compared to 56/60 versus 60/60 at `seed_base=0`). The exact `successes` counts and the exact magnitude of the top-end decline are not claimed to generalize beyond the pinned numbers at each seed base; it is the *shape* — threshold, then plateau, then decline — that replicates. The additional `seed_base=5000` spot-check remains an informal, ungated note (reproduced the same qualitative shape when checked by hand, but is not bound to a regression test). For the heterogeneity sweep, the cross-seed replication is likewise gated: an independent `seed_base=7000` run (a seed block sharing zero seeds with the `seed_base=0` sweep) is a **gated** regression test (`test_heterogeneity_strict_ordering_replicates_at_a_disjoint_seed_base`). At that disjoint block the strict ordering reproduces with successes 60/54/14/5 (tight/medium/wide/very-wide) — different exact counts from the `seed_base=0` run's 60/56/15/2, as expected for a different seed block, but the identical monotonic-decrease shape. The exact `successes` counts at any given seed base are not claimed to generalize beyond the pinned numbers; it is the *ordering* that replicates. The mechanistic account offered for Experiment A was originally inference from trace inspection alone, explicitly flagged as distinct from the (better-supported) claim that decay has a real, non-monotonic causal effect on the rate within this controlled simulation; it is now additionally corroborated by a dedicated `sensed_concentration_cap` ablation (above) showing that capping alone recovers convergence at decay $\in \{0.10, 0.30\}$ — sufficiency evidence for the account, not a full mechanistic proof (see that section's "What this does and does not establish"). Nothing here establishes these effects analytically (no closed-form model of convergence rate as a function of decay or heterogeneity width is derived or claimed) — every number in this section is an empirical measurement from a real, seeded simulation run, not a theoretical prediction.

Why no multiple-comparisons correction (Bonferroni/Holm) is applied across these sweeps. The decay sweep has six points and the heterogeneity sweep has four; eyeballing several points across a sweep for "the interesting difference" is exactly the kind of selective-inference exposure a multiple-comparisons correction exists to guard against, so it is worth stating plainly why none is applied here rather than leaving the omission implicit. First, each sweep's *primary* whole-sequence claim rests on a single, pre-specified omnibus test — one Cochran–Armitage trend test per sweep (`cochran_armitage_trend_test`, $Z = +14.57$ for decay, $Z = -12.76$ for heterogeneity, both $p < 10^{-10}$) — not on scanning the sweep's points pairwise for the smallest p-value. A single test per sweep has no multiple-comparisons problem to correct for. Second, the one genuinely exploratory *pairwise* comparison this manuscript does report and interpret — the decay $\{0.60, 0.80\}$-versus-1.00 dip, via `fisher_exact_test_two_sided` — is already disclosed above as **non-significant** ($p = 0.1187$, not clearing $\alpha = 0.05$) and is explicitly not used to support a positive claim on its own: its evidentiary weight is stated to rest on the sweep's overall shape and its replication at disjoint seed bases, not on that one comparison clearing a significance threshold. A formal correction exists to prevent an uncorrected small p-value from a scan of many comparisons being over-interpreted as significant; that failure mode requires actually reporting an uncorrected pairwise p-value as if it, alone, established significance. This manuscript does not do that anywhere: the only pairwise test it reports is disclosed as failing to reach significance, and the only comparisons it treats as decisive are Wilson-interval-overlap checks between two pre-specified conditions (e.g. Experiment B's real-mechanism- versus-null and full-versus-zero-deposit comparisons), not a multiple-testing search over many candidate pairs for the one that clears a threshold. A correction is the right tool for a different failure mode than the one this manuscript's reporting practice actually exhibits.

6.6 The optional formal side-spec: shipped, not cut

Per ISC-35/36, this template ships both a Lean 4 model (`formal/lean/AntProtocol.lean`) and a TLA+ specification (`formal/tla/AntProtocol.tla` + `AntProtocol.cfg`) of the handshake protocol, each wired to exactly one real runnable check inside `scripts/check_formal_specs.sh` — fresh pinned Lean elaboration and axiom audits, and the TLC model checker (`tla2tools.jar`) for the TLA+ spec. Both checks are optional (they require `lake`/`elan` and a Java runtime respectively, and are excluded from the core pipeline; full unpaid acceptance executes them with explicit pinned acquisition and a shared deadline), but neither is decorative: the script fails non-zero if either check fails, and both have been executed at least once with captured output (recorded in `ISA.md`'s `## Decisions`). This is a deliberately different evidentiary register from the rest of this paper's claims — Lean/TLA+ model the *protocol design*, independent of whether this template's Python implementation actually conforms to that design; no claim in this paper conflates a passing Lean/TLA+ check with a proof about the Python code in `src/`.

6.7 Formal side-spec expansion: what grew, and what still needs wiring

Theorem 15 (Mechanically verified protocol invariants)**.** *Three claims about the handshake protocol's* design *(not this template's Python implementation, per the scoping note above) are machine-checked, zero-*`sorry`*, zero-extra-axiom results, re-verified end to end by* `scripts/check_formal_specs.sh`*: (i)* `step_to_closed_cases` *(Lean 4) — every transition into the* `closed` *phase originates from* `idle`*,*

established, or handshaking, never any other phase; (ii) cannot_reuse_consumed_token (Lean 4) — a second, un-reassigned use call on a consumed session token has an unsatisfiable precondition; (iii) NoFalseEstablishment (TLA+, AntProtocolFaulty.tla) — a peer is never established without the other peer having genuinely sent *the corresponding real message at some point (*send-provenance*), checked over 494 generated / 92 distinct states under arbitrary drop/corrupt/duplicate fault interleaving, and independently proven non-vacuous by a permanent negative control (below) rather than merely asserted.*

Both formal side-specs grew since the "shipped, not cut" decision above was first recorded, and both expansions were re-verified as real, runnable checks rather than taken on faith.

A RedTeam pass found, and a mutation test confirmed, that clause (iii) above originally overclaimed what NoFalseEstablishment actually guards against. The invariant is built on iSentHello/rSentAck — pure send-side history flags, set once by a genuine SendHello/accept-branch and never reset. That makes the invariant a check of **send-provenance** ("was a real message genuinely sent by the other peer, at some point"), not of **content-integrity resilience** ("was the specific delivered frame uncorrupted"). A direct mutation test made this concrete: widening ReceiveHello's guard from m.kind = "hello" to also accept a "corrupt"-tagged frame enlarges the reachable state space (92 → 136 distinct states) yet NoFalseEstablishment still reports zero violations — because Corrupt preserves a message's sender field and only ever acts on a frame a genuine SendHello already placed on the channel, so accepting a corrupted-but-genuinely-originated frame still satisfies send-provenance. This is exactly the same *near-vacuity* failure class already found and fixed twice on the Lean side (no_direct_idle_to_established, closed_only_via_known_paths) — an invariant that holds regardless of whether the mechanism it is supposed to gate is correctly implemented.

The fix mirrors this template's own tests/mypy_fixtures/bad_*.py negative-control discipline rather than merely reworking the prose: formal/tla/AntProtocolFaultyNegControl.tla is a permanent, DELIBERATELY BROKEN sibling model — verbatim copy of AntProtocolFaulty.tla plus one added action, ForgeHello, which injects a [kind |-> "hello", sender |-> "I"] message onto the channel *without* setting iSentHello (an unauthenticated/spoofed hello the initiator never actually sent). scripts/check_formal_specs.sh runs TLC against it with **inverted** pass/fail logic — a reported violation is the expected, correct outcome; a clean "no error found" would mean the negative control itself is vacuous. Re-verified: TLC finds the violation in a 3-state trace (Init → ForgeHello → ReceiveHello establishes rPhase while iSentHello is still FALSE), proving NoFalseEstablishment genuinely depends on send-provenance holding, and clause (iii) above is now worded to the guarantee actually established rather than the broader one the original phrasing implied.

Lean 4 (formal/lean/AntProtocol.lean) grew from 3 theorems to 7 audited property theorems (plus 3 reachability-witness theorems used as their preconditions, 10 declarations total), all zero-sorry, and — per the file's own #print axioms lines, re-confirmed by a clean lake build — depending on no axioms beyond Lean's own kernel: phase_exhaustive (the four Phase constructors are the only inhabitants of the type — a background exhaustiveness fact the earlier revision left implicit), closed_is_terminal (no Step relation has closed as its source — the direct Lean counterpart of "no outgoing edges from the ClosedSession phase" in protocol/session.py), a minimal affine SessionToken/use model plus cannot_reuse_consumed_token (the precondition a second, un-reassigned use call would need is unsatisfiable — the first Lean counterpart to the runtime SessionConsumedError discipline the type system cannot close, per sec. 3), and no_direct_idle_to_established — which *replaces* the prior no_skip_to_established, discarded because it reduced to a restatement of established_requires_handshaking via a vacuously-true Reaches idle idle conjunct rather than proving the one-step non-existence claim its name promised. Re-verified directly: lake build → Build completed successfully (3 jobs), with all 7 #print axioms lines confirming zero extra axioms.

A second cross-vendor audit pass caught one more instance of the exact vacuity class no_direct_idle_to_established had already been introduced to fix: the theorem closed_only_via_known_paths concluded a three-way Reaches idle established ∨ Reaches idle idle ∨ Reaches idle handshaking disjunction, and Reaches idle idle is unconditionally provable via Reaches.refl regardless of which run was passed in — so the whole disjunction was satisfiable by a fixed proof term that never inspected its hypothesis. Rewording the conclusion to reference the run's own extracted predecessor phase did not escape the trap either, because in this small, 4-phase state machine every candidate predecessor is trivially reachable from idle by *some* path independent of the specific run — the Reaches-wrapped framing is inherently satisfiable regardless of hypothesis in this model. The theorem was removed rather than patched a second time; the genuinely load-bearing, falsifiable content it was built on top of — step_to_closed_cases, a direct, universally-quantified claim about the Step relation itself ("every transition into closed comes from idle, established, or handshaking — never anything else"), false if a hypothetical sixth constructor targeted closed from elsewhere — was promoted to a named, #print axioms-checked theorem in its place. The audited theorem count stays at 7 (10 declarations including the 3 reachability witnesses); the set of what is proved changed to remove the vacuous member.

TLA+ gained two things, not one. First, AntProtocol.tla added a liveness property, HandshakeEventuallyResolves (□(handshaking ⇒ ◇(established ∨ closed))), which only holds relative to the weak-fairness conjuncts (WF_vars(Ack), WF_vars(AbortHs)) added to Spec — without them, TLC's exists-a-behavior semantics would permit a "coward" run that stalls in handshaking forever via stuttering, making the property vacuously unprovable. Re-verified: TLC's temporal-property check over the complete state space (5 distinct states) reports **"Model checking completed. No error has been found."** Second, a genuinely new model shipped alongside the original: AntProtocolFaulty.tla (+ .cfg) is a two-peer (initiator/responder) model with an explicit message channel as first-class state — not a copy of the single-peer AntProtocol.tla — with the Send, ReceiveHello, ReceiveAck, Corrupt, DeliverCorrupt, and Duplicate actions mirroring network/bus.py's real drop/corrupt/duplicate fault modes, checked against NoFalseEstablishment (a peer is never established without the other peer's real message having actually been sent, even under arbitrary fault interleaving — the

direct TLA+ analogue of ISC-23/24). Re-verified: TLC reports **"Model checking completed. No error has been found"** over 494 states generated, 92 distinct states.

Update — now wired. `scripts/check_formal_specs.sh` was extended to run TLC a second time, against `AntProtocolFaulty.cfg/AntProtocolFaulty.tla`, immediately after the original `AntProtocol` check, and now a *third* time against the negative control `AntProtocolFaultyNegControl.cfg/.tla` described above (with inverted pass/fail logic) — all three TLC runs plus the Lean build must pass for the script to exit zero. Re-verified end to end: `lake build` (7 audited + 3 witness theorems, zero `sorry`) → `PASS`; TLC on `AntProtocol.tla` (liveness included) → `PASS`; TLC on `AntProtocolFaulty.tla` (494 states generated, 92 distinct) → `PASS`; TLC on `AntProtocolFaultyNegControl.tla` → violation correctly detected in a 3-state trace → `PASS`; `check_formal_specs.sh: both formal side-specs passed`, exit 0. All four checks now run under the same single automated command, closing the ISC-36 anti-criterion gap this section previously flagged as open — and, per the negative-control addition above, this section no longer merely asserts `NoFalseEstablishment` is non-vacuous, it demonstrates it the same way `tests/mypy_fixtures/` demonstrates the mypy oracle is non-vacuous.

6.8 Threats to the claims, honestly stated

Nothing in this paper should be read as a claim that `mypy --strict` detects every illegal-state bug a determined adversary could construct in this codebase, nor that the fault-injection suite's four modes (drop/reorder/duplicate/corrupt) exhaust the space of network faults a production system would face — this template explicitly excludes real sockets, multi-process execution, and a message broker (`Out of Scope`). The colony integration test's stigmergic convergence result is a genuine, reproducible property of the specific three-agent, two-location, five-tick configuration tested; it is not evidence about convergence at colony scales, location counts, or tick horizons this template does not test. **The same caveat applies, separately, to the $N = 150$ statistical claim above** (a RedTeam pass flagged that an earlier revision of this section scoped the caveat to the three-agent demo test only, leaving the headline 8-agent/2-location/30-tick statistical claim with no disclosed generalization limit of its own): "140/150 converge, Wilson lower bound 0.8816" is a property of *that* configuration, not a general law about stigmergic consensus at other agent counts, location counts, or tick horizons. Widening any of these claims would require the same discipline this paper asks of every claim already in it: a new negative-control test or fault-injected scenario before the wider claim is made, not merely a wider sentence in the manuscript. The pre-registered experiments above narrow this further, not widen it: they fix `num_agents=8`, two `locations`, and `num_ticks=30` throughout, and each experiment varies *exactly one* dimension (decay, mechanism-vs-random, heterogeneity width, or — for the zero-deposit ablation closing Experiment B's stigmergy- attribution confound — `deposit_amount` alone) while holding the others at the calibrated baseline. None of them licenses a claim about how decay, the mechanism's advantage over chance, heterogeneity, or the deposit-channel attribution would behave at a different colony size, location count, or tick horizon — only cross-seed robustness supports the narrower claim that the *qualitative shape* observed at `seed_base=0` is not an artifact of that one specific seed sequence, and that support is uneven by design: the heterogeneity ordering's replication at the disjoint `seed_base=7000` is now a gated regression test, whereas the decay threshold, the real-vs-null gap, and the zero-deposit ablation are each only established at `seed_base=0` (the decay threshold has additional informal spot-checks at other seed bases; the real-vs-null gap and the zero-deposit ablation do not, and neither is gated by a cross-seed-base regression test the way the heterogeneity ordering is).

7 References

Bibliography lives in `manuscript/references.bib` and is read by Pandoc during PDF render. The build pipeline invokes Pandoc with `--natbib`, so every `[@key]` citation in the manuscript is rewritten to the appropriate `\cite{}`/`\citep{}`/`\citet{}` LaTeX command and resolved against the bib file.

To validate that `references.bib` is syntactically clean and contains the required fields per entry type:

```
uv run python -m infrastructure.reference.citation.cli validate \
    projects/templates/template_formal/manuscript/references.bib --strict
```


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